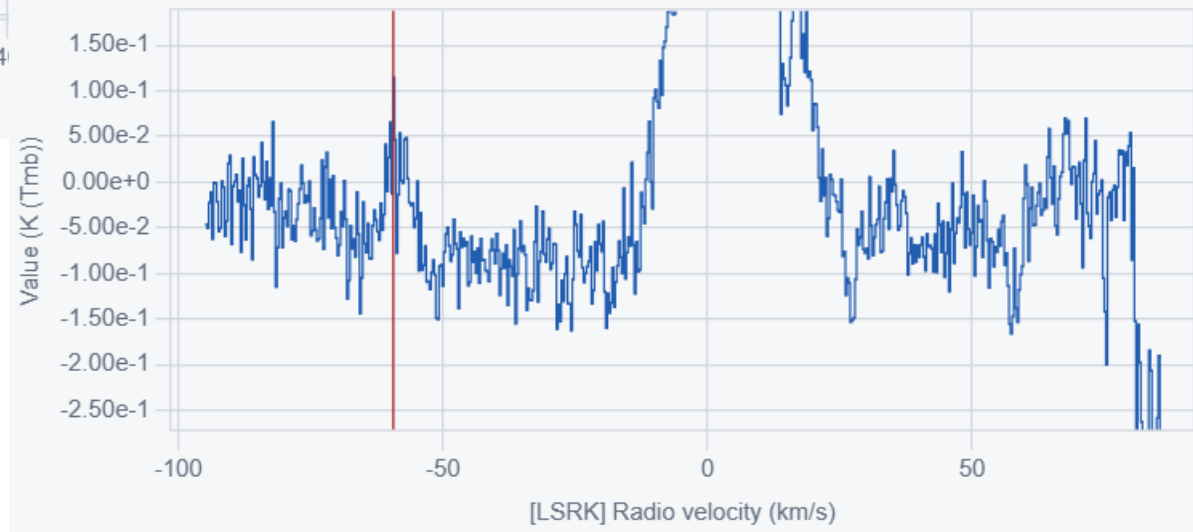
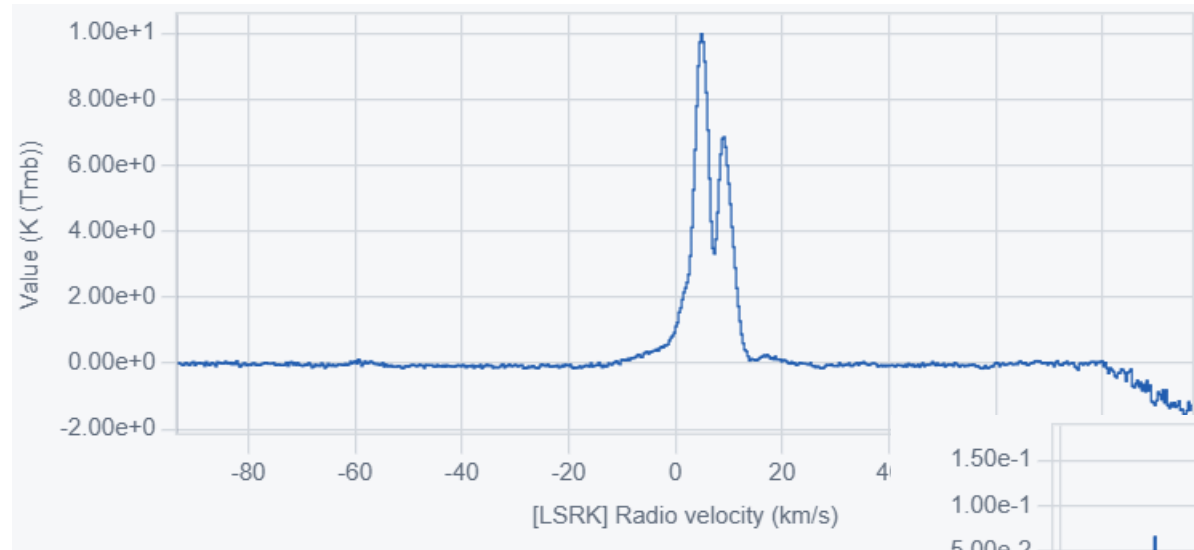


9 April 2026

<https://github.com/Lim1029/W40-CII-Analysis/blob/main/>

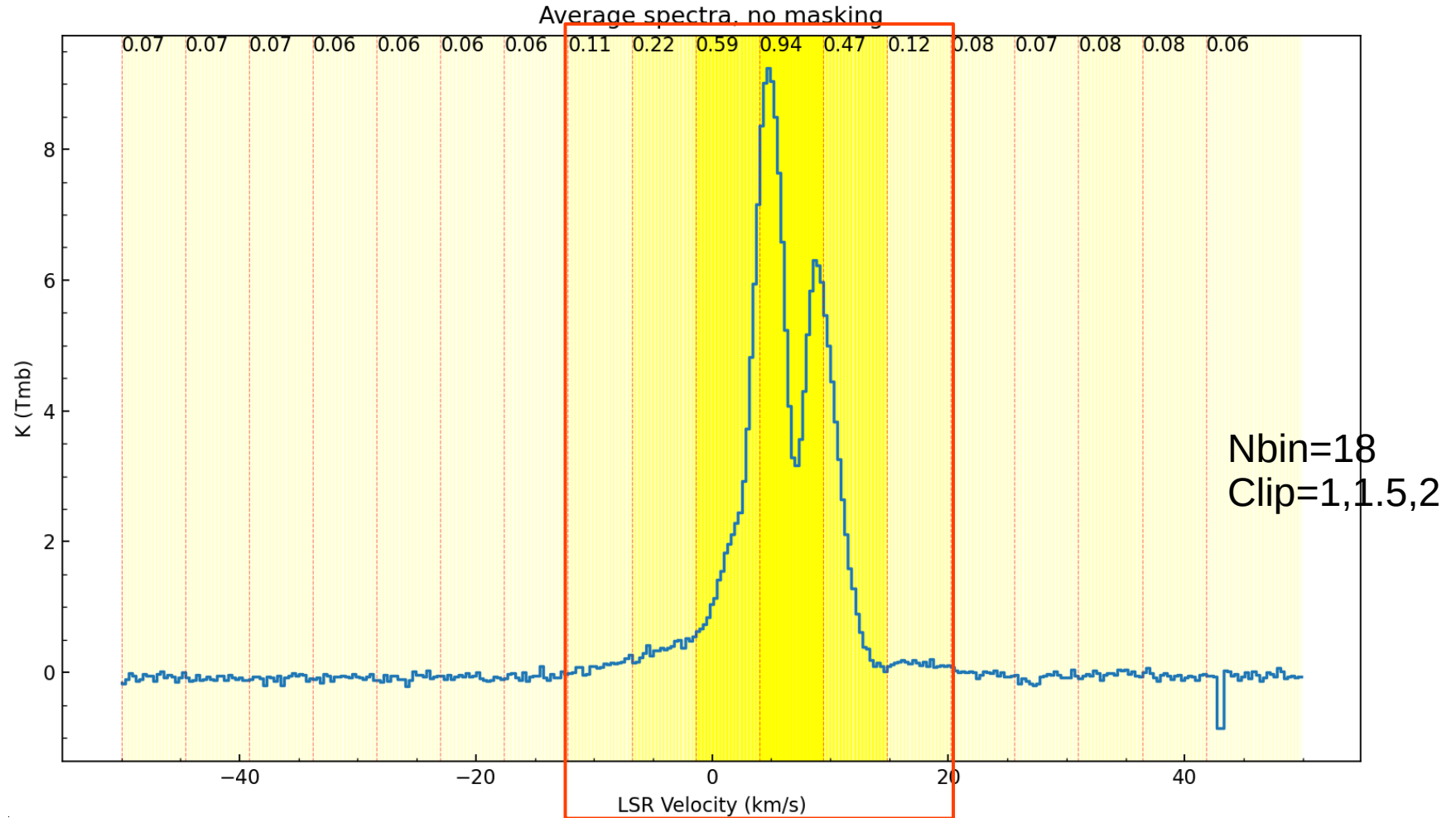
Average Spectrum

visual inspection on average spectrum does not show faint features well



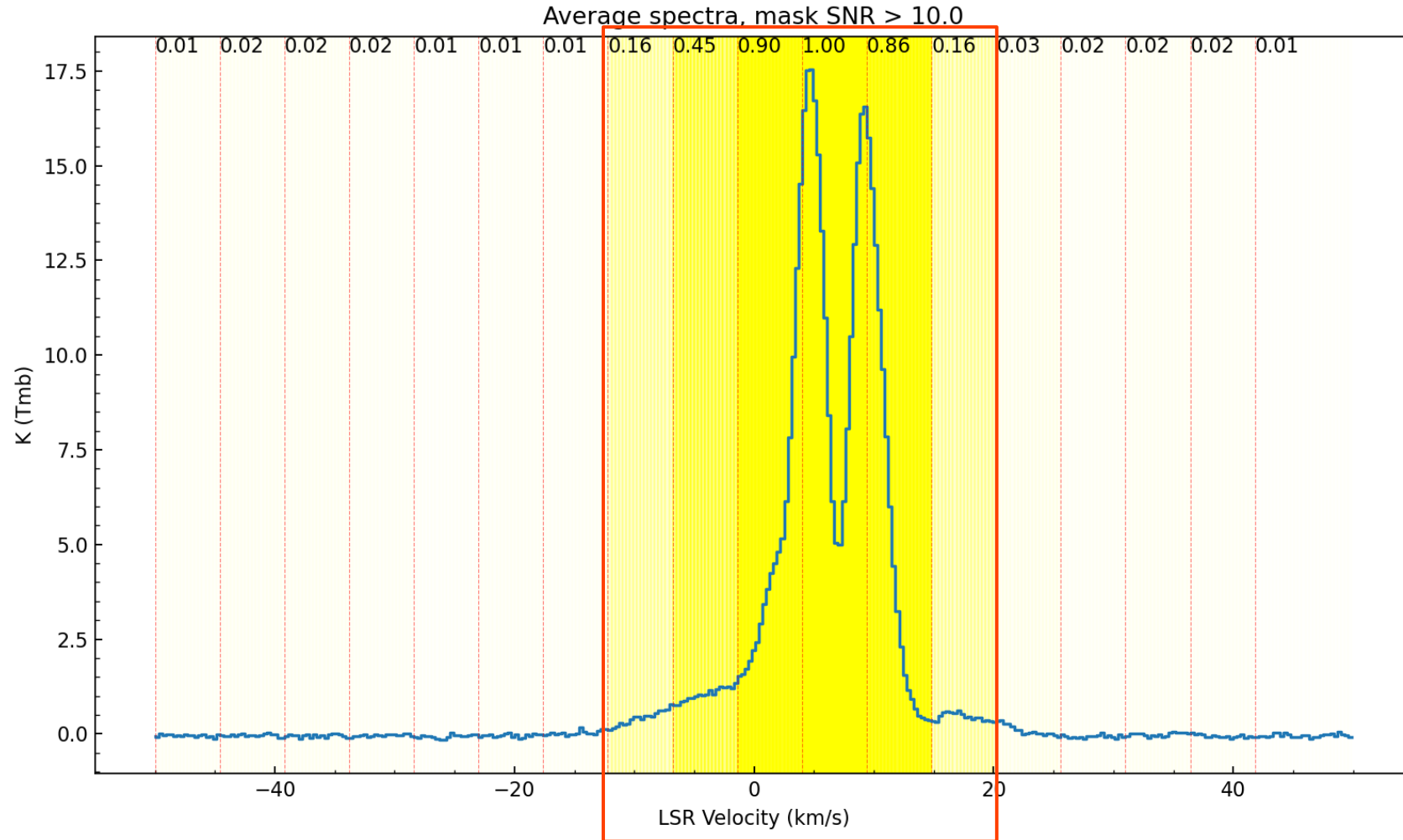
Dynamic baseline windowing

selecting the range $[-12, 20]$ km/s; https://lim1029.github.io/jcmt_window_demo.pdf



Dynamic baseline windowing

selecting the range $[-12, 20]$ km/s; https://lim1029.github.io/jcmt_window_demo.pdf

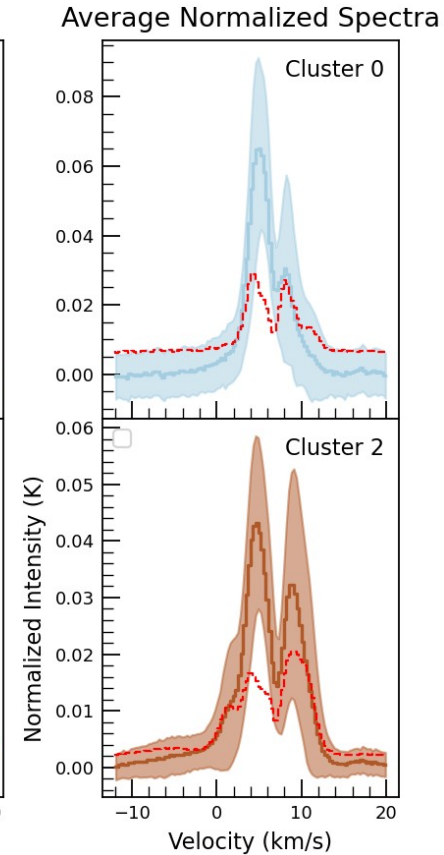
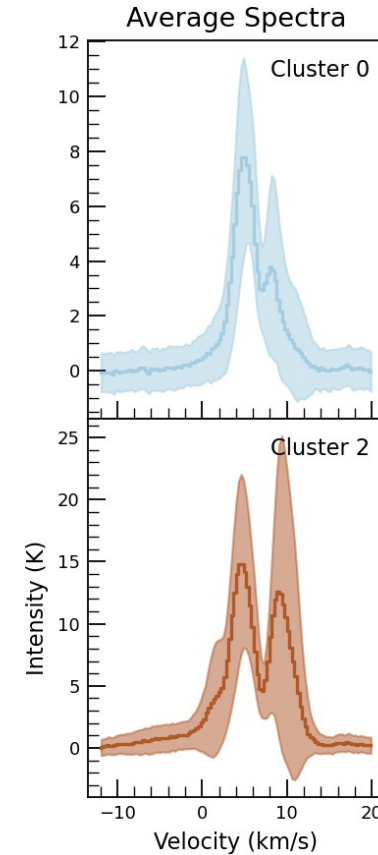
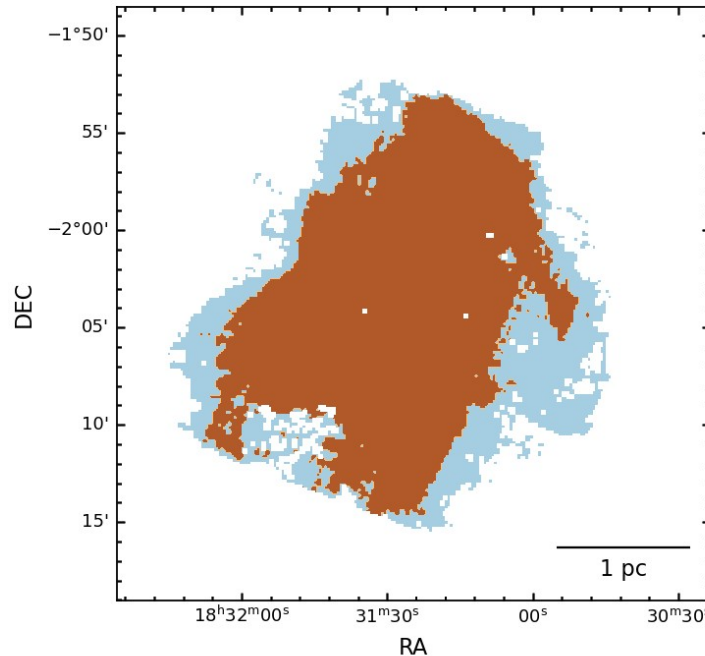


Nbin=18
Clip=1,1.5,2

Selecting the best number of clusters (ncomp): Method 1

By selecting the ncomp with results in the lowest BIC. Max ncomp = 30

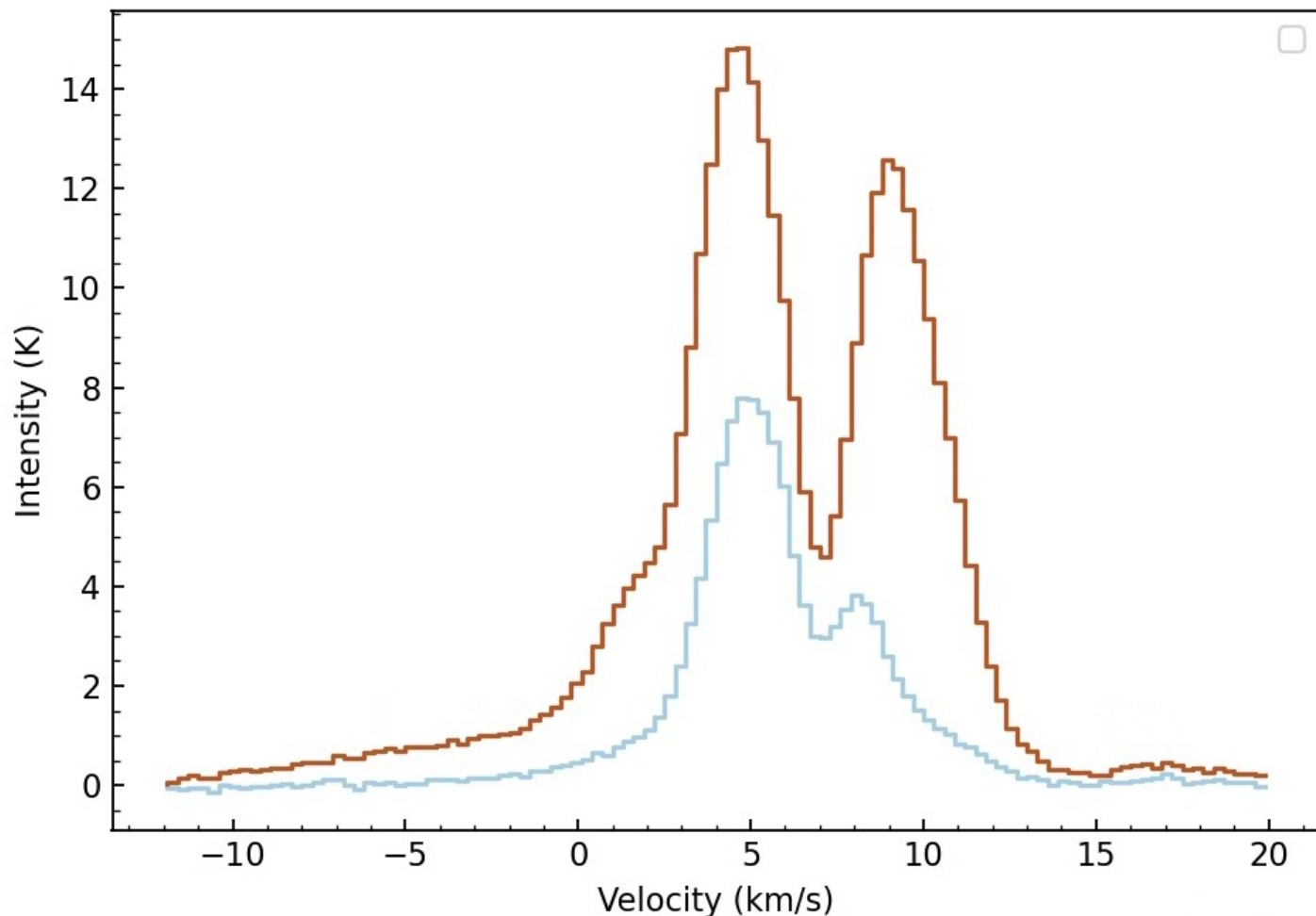
Number of components	BIC score
2	3 -1.173089e+07
3	4 -1.170822e+07
4	5 -1.164567e+07
1	2 -1.159345e+07
5	6 -1.158100e+07



Ncomp=3 (2 components + 1 blank) returns the lowest BIC

(Cont) Selecting the best number of clusters (ncomp): Method 1

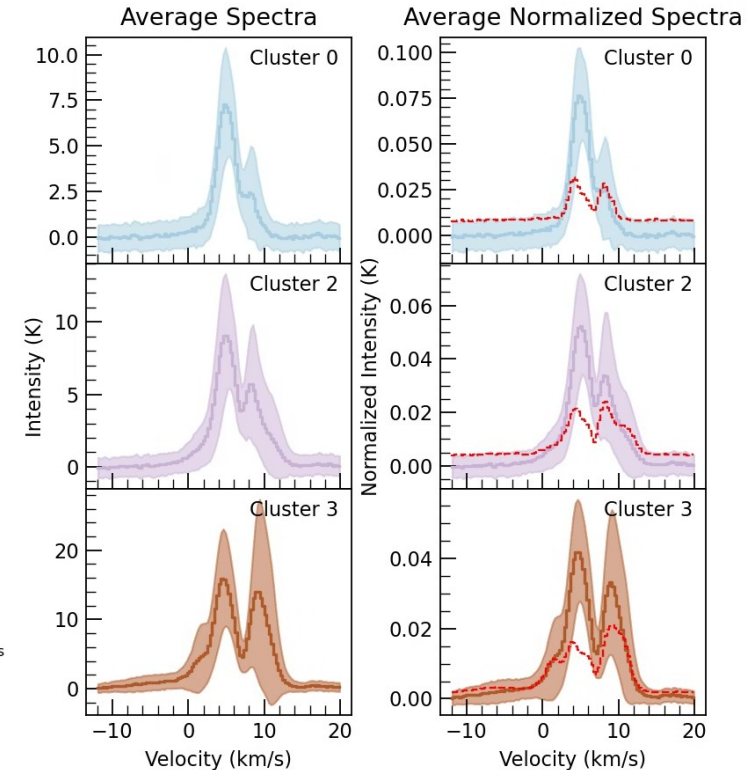
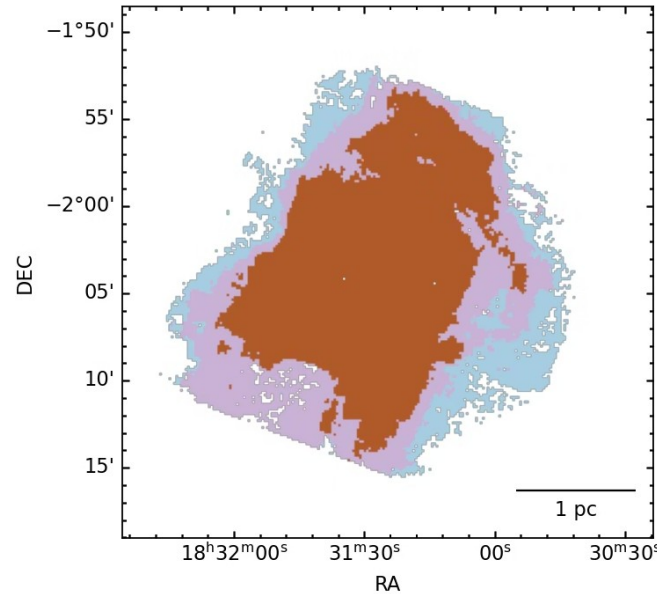
By selecting the ncomp with results in the lowest BIC. Max ncomp = 30



Selecting the best number of clusters (ncomp): Method 2

By selecting the ncomp with results in the lowest AIC. Max ncomp = 30

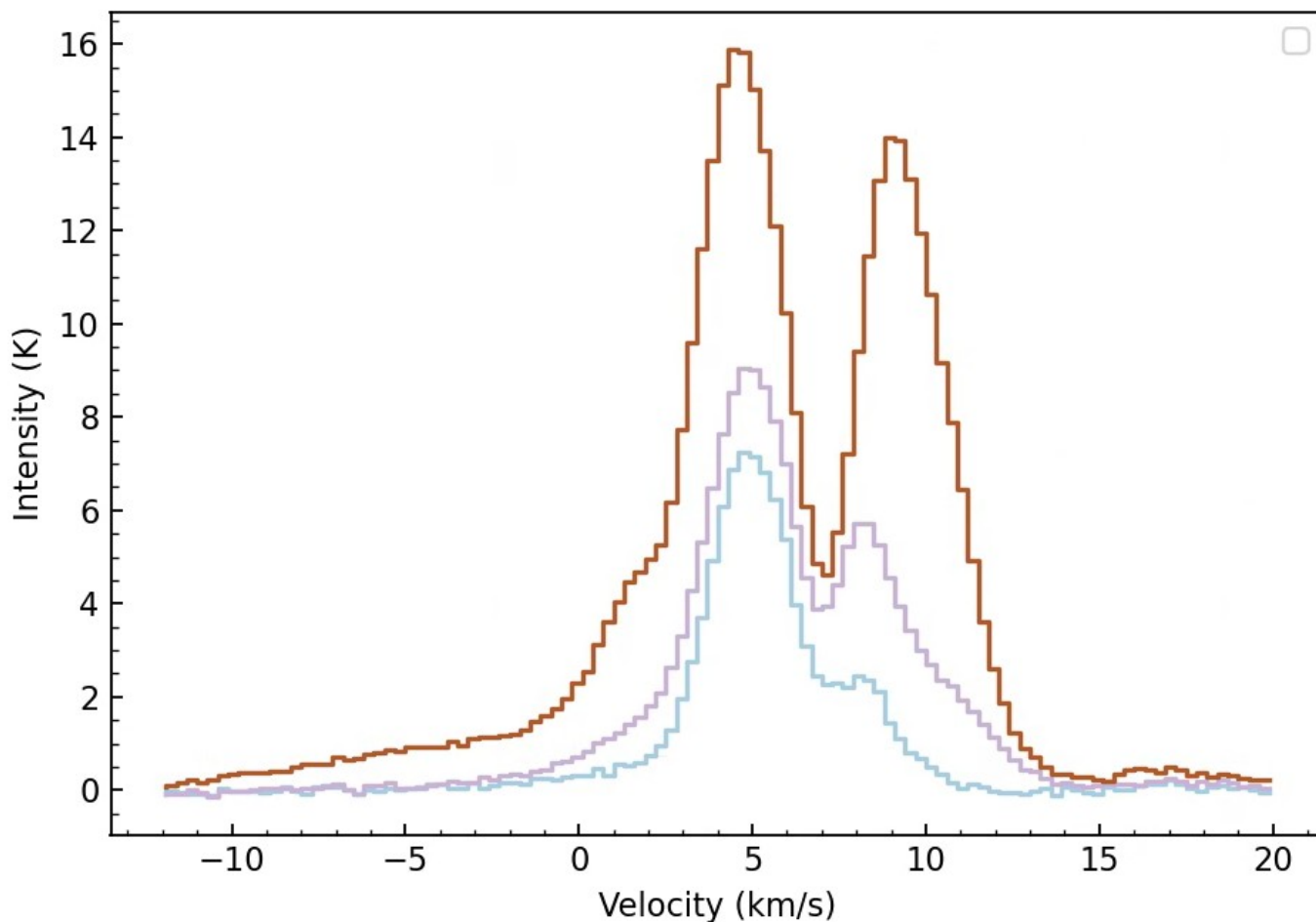
Number of components	AIC score
3	4 -1.187904e+07
4	5 -1.185930e+07
2	3 -1.185904e+07
5	6 -1.184597e+07
6	7 -1.182728e+07
7	8 -1.180481e+07
8	9 -1.178918e+07



Ncomp=4 (3 components + 1 blank) returns the lowest AIC

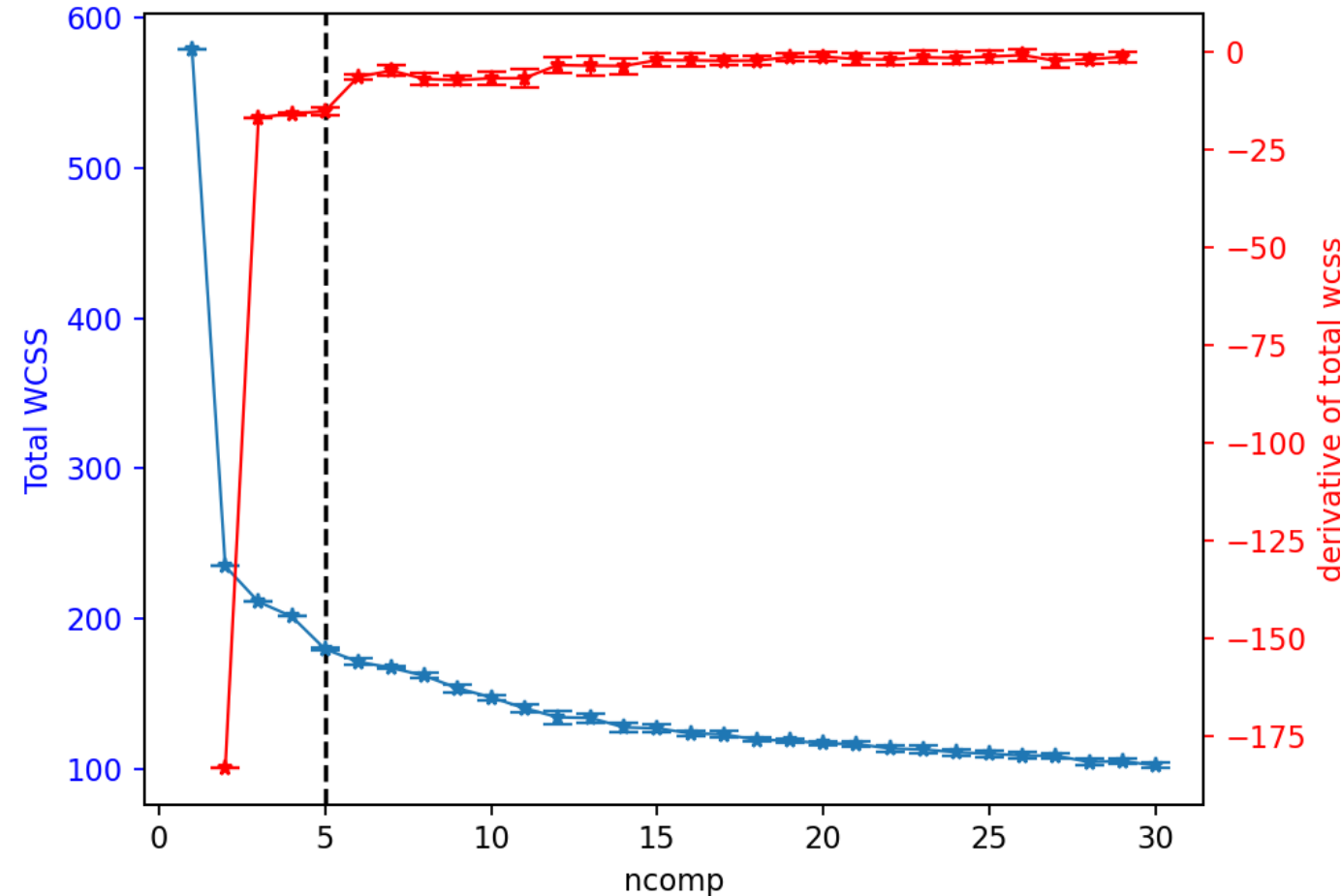
(Cont) Selecting the best number of clusters (ncomp): Method 2

By selecting the ncomp with results in the lowest AIC. Max ncomp = 30



Selecting the best number of clusters (ncomp): Method 3

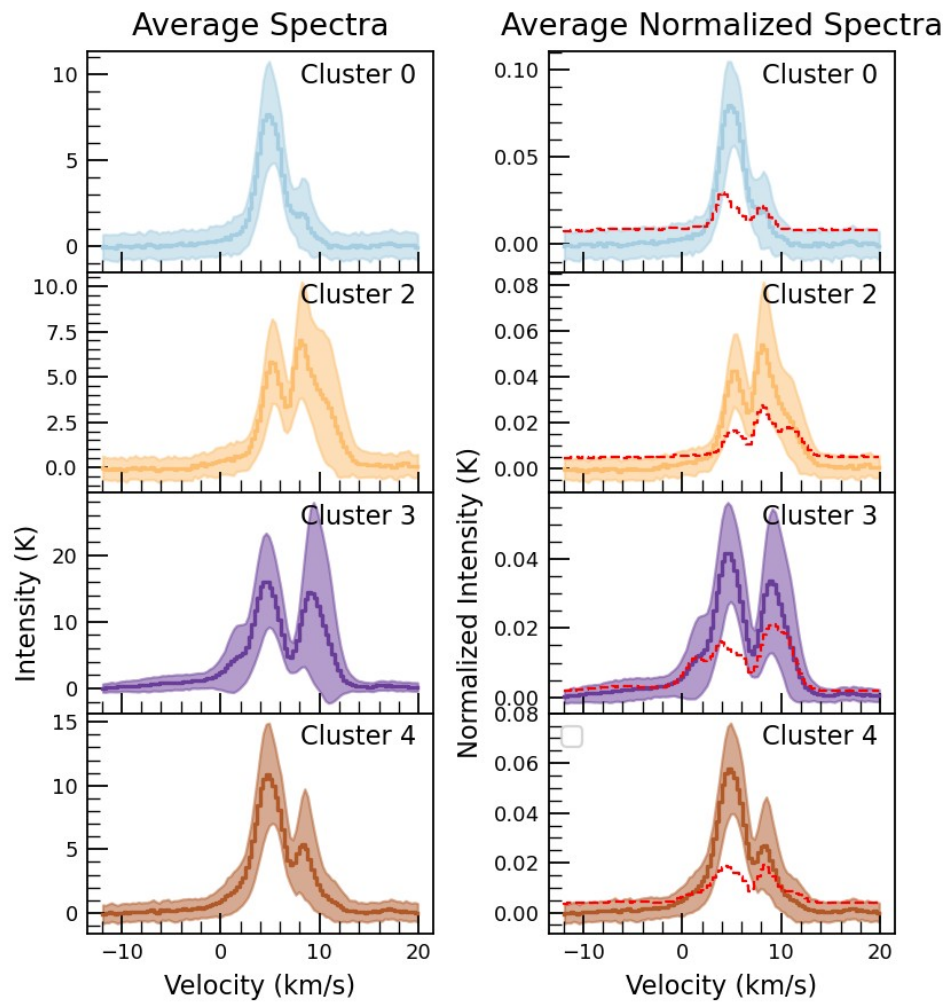
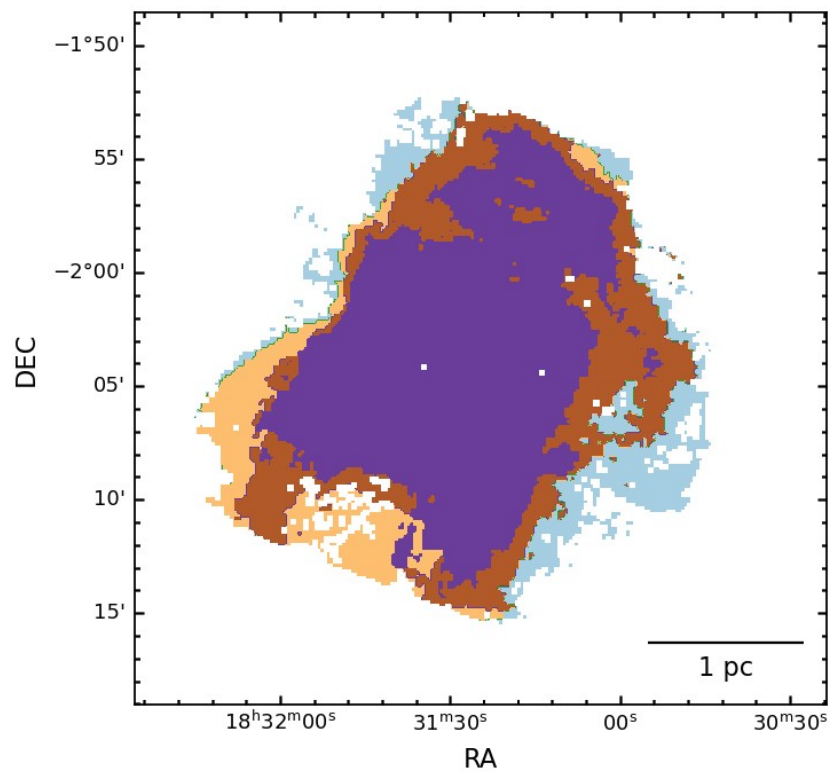
By minimising the uncertainty of normalised averaged spectra



Kneedle algorithm returns knee at ncomp=5 (black vertical dashed line)

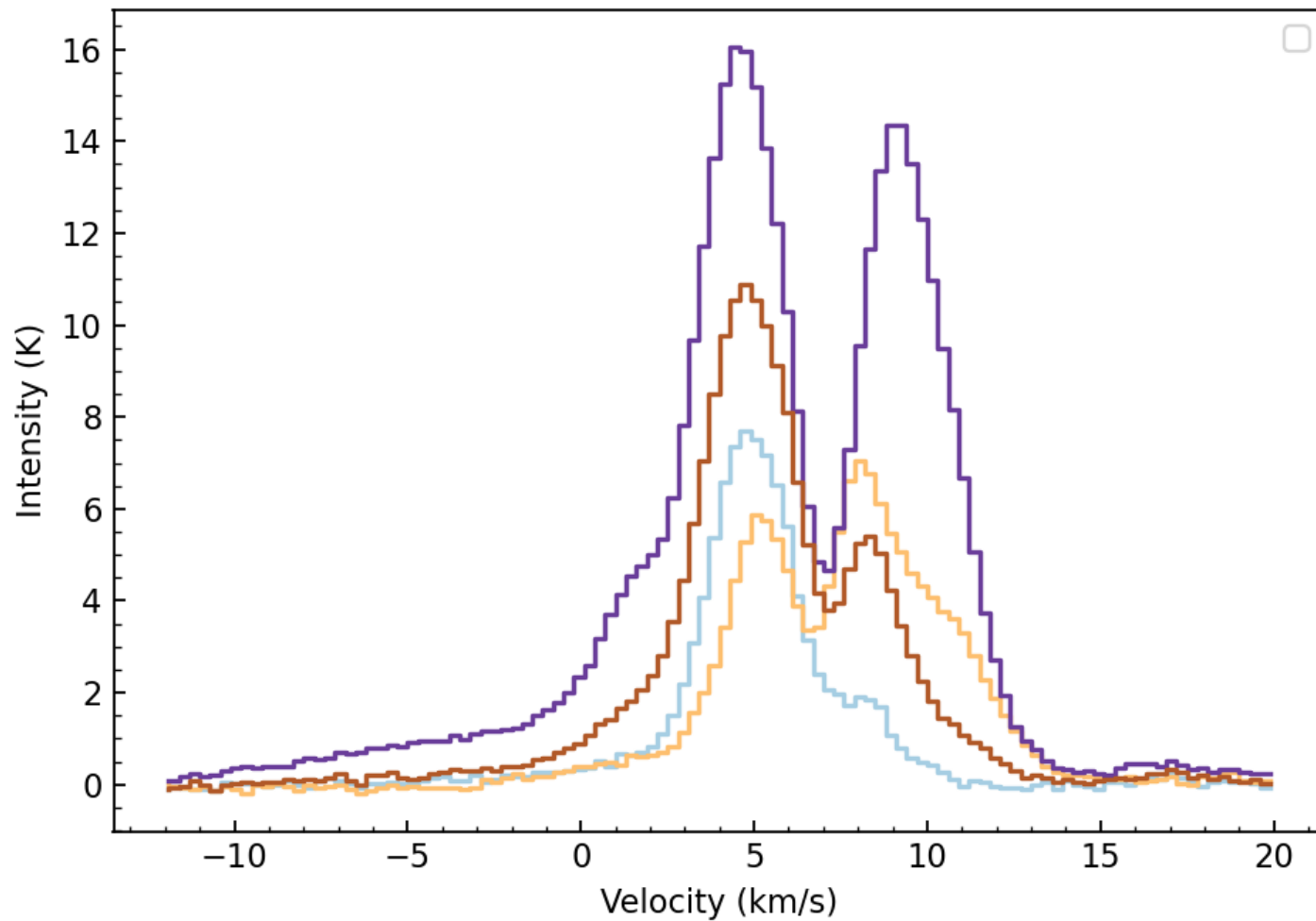
*Uncertainty of each ncomp is from its 10 times run standard deviation

Running with ncomp=5



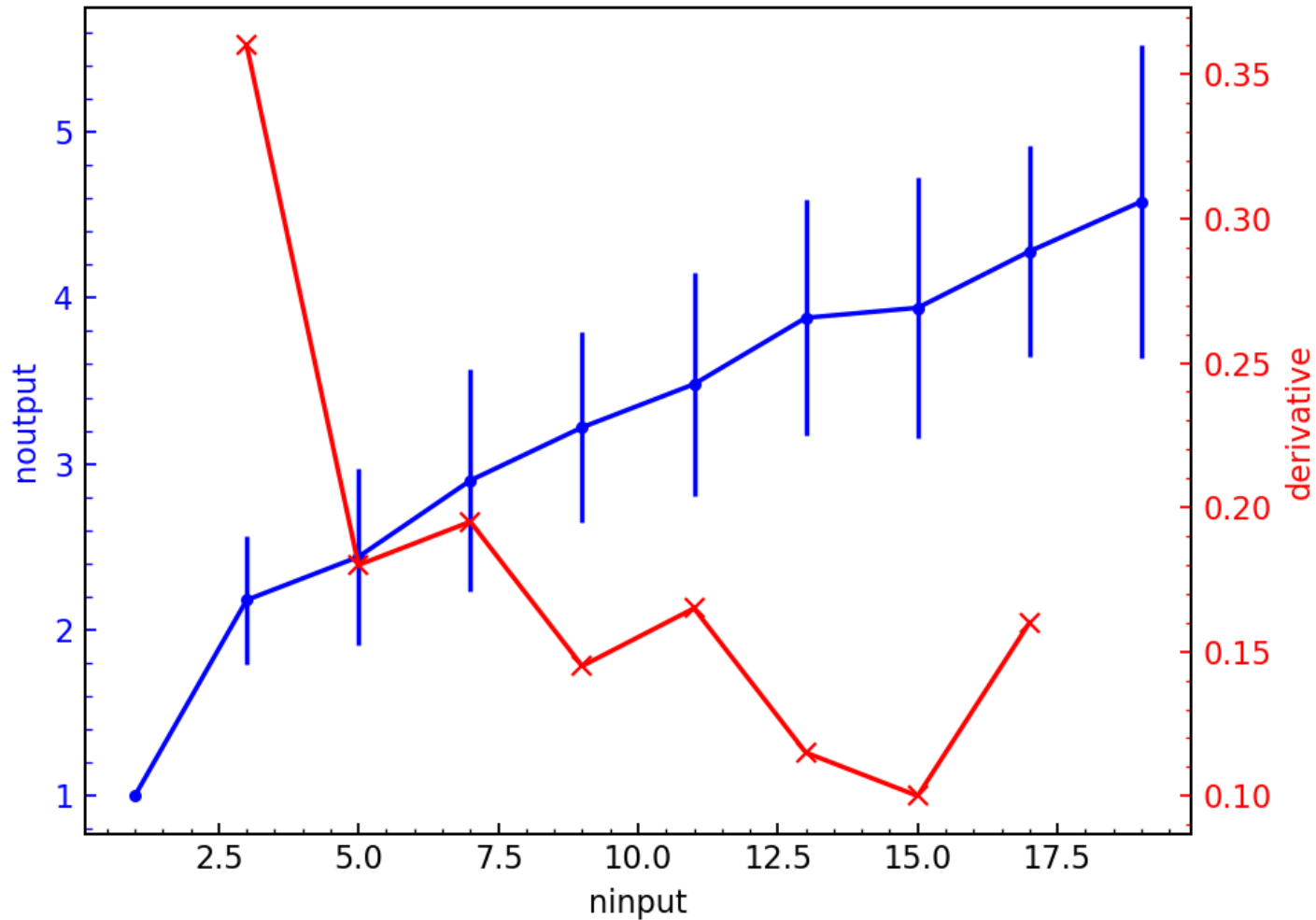
*red dotted line (and filled colours) indicates standard deviation

Running with ncomp=5



Running with PyGMMis code

Source: Tiwari+23



First breaking point
(minimum derivative) is
ninput ~ 15, noutput=3.9

default parameters

Run PyGMMis with ninput=15

Source: Tiwari+23

